SEPTEMBER REVENUE, SAHYADRI STORES

Rs ~~24,03,00,20~~

Our agent said this with full confidence.

The real number is **Rs 2,40,300**. No error. No warning.

How do you catch an AI agent that is wrong and doesn't know it?

THE PROBLEM

Agents fail in three ways. Only one of them is easy to see.

LOUD

The tool throws an error

"no such column: amount". Easy to catch, easy to retry.

QUIET

The number is just wrong

Prices read in paise, a missing GROUP BY. The query runs fine.

SPREADING

The wrong number is reused

The next question builds on it, and the mistake travels.

QUERYMEND

Ask your shop database a question in plain English. If the answer is wrong, the agent notices, fixes it, or says it can't.

qwen2.5:3b · LangGraph · SQLite · Chroma · runs fully offline, free

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Topic 17: an agent that detects and recovers from its own tool failures

PBA 1

One agent that notices its own tool failures

agent
qwen2.5:3b with run_sql

↓ tool call

tools
read-only SQL, clear errors

↓

route_after_tools
did the tool fail?

↓ error

handle_failure
typed hint, 3 tries max

SLIDE VERSION

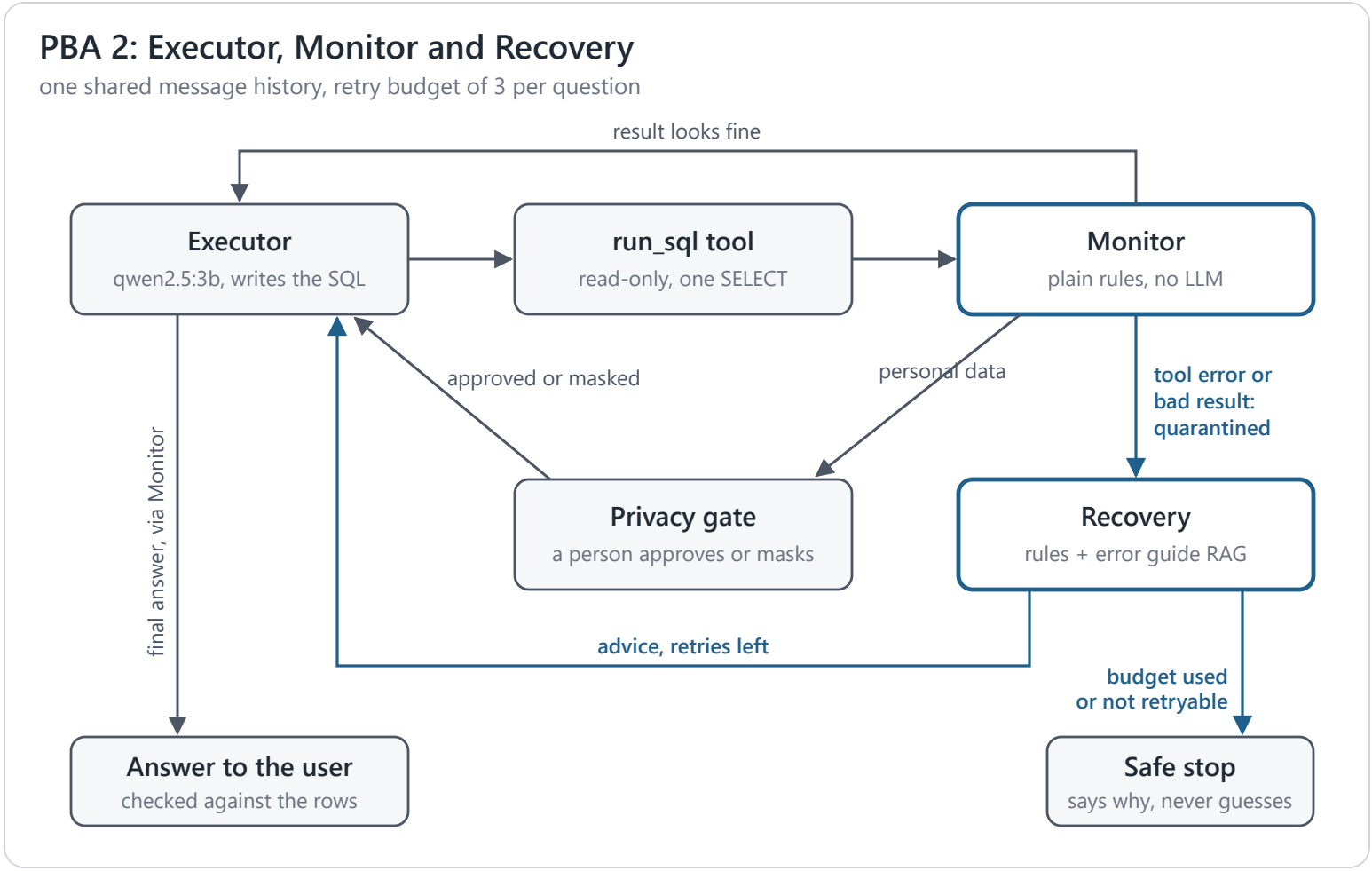
One generic notice for every error. Sent the same broken query 7 times. Told the user to run a DELETE by hand.

OUR REFINEMENT

Sort the error, give a specific hint, cap retries at 3, lock the tool when retrying can't help.

Still blind to wrong data that raises no error.

Three agents, one job each



Executor writes and runs the SQL. The only one that touches the data.

Monitor checks every result and answer with plain rules. No model, same verdict every time.

Recovery gives advice from our error guide, or stops after 3 retries.

UNIT 3 LAYER

It remembers, it resumes, and it reads its own error guide

CHECKPOINTER

Follow-ups work

"And in September?" knows what you asked before. A crashed run resumes.

STORE

Your words stick

"Premium product means above Rs 1500" survives a restart.

FAILURE LOG

Nothing is hidden

Every failure, its SQL and what happened next, in SQLite.

ERROR GUIDE RAG

Specific advice

15 sections in Chroma. Says "I don't know" when the guide doesn't cover it.

LIVE DEMO

Four runs, about three minutes

1. One corrupted result, Monitor off, then on
   `python ta2/run.py containment`
2. Five injected failure types and the evidence table
   `python ta2/run.py failures`
3. Personal data paused for a person: approve, then reject
   `python ta2/run.py privacy`
4. Your question, or your edge case
   Every run is also saved in `traces/` as a backup

ERROR PROPAGATION, 10 RUNS EACH

With the Monitor on, a bad number never reaches the next question

MONITOR OFF

10/10

runs carried the wrong number into the follow-up answer

MONITOR ON

0/10

The bad result is swapped for a quarantine note before any agent reads it

Cost: 2 extra model calls and about 2 seconds per question.

SAME 5 CASES, 3 RUNS EACH

Fewer wrong answers, more model calls

	PBA 1	PBA 2
Wrong answers given to the user	4 / 15	**0 / 15**
Silent corruption caught	0 / 3	**3 / 3**
Model calls, corrupt case	2.0	6.0
Model calls, clean case	2.0	3.0

WHAT GOT WORSE

Recovery costs 2 to 3 times the calls. Even the clean case needs one retry.

FOUND BY TESTING

A missing GROUP BY slipped past our number checks. The Monitor now reads the SQL too.

WHAT'S NEXT

From a course project to a tool a shop owner can trust

1. Owners set their own limits
   "An order is never above Rs 50,000"
2. Real databases through a read-only user
   Postgres or MySQL, same checks
3. A simple page: the answer, the SQL, and a note when the Monitor stepped in

github.com/Swaraj-Mandre/FailureDetectionAgent · Thank you. Questions?